Observing the Galactic HI 21-cm Line with the Big Horn Receiver

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ABSTRACT

We present a complete calibration-first analysis of the Big Horn 21-cm receiver system on the roof of New Campbell Hall, spanning equipment characterization, receiver noise temperature measurement, and Galactic HI line observations. Coaxial cable attenuation was measured as $\alpha = 0.73 \pm 0.13\,\mathrm{dB/m}$ (SDR) and $0.80 \pm 0.03\,\mathrm{dB/m}$ (power-meter), with the unknown roof–lab cable length inferred as ≈ 36–$40\,\mathrm{m}$. Transmission-line reflectometry gave a coax propagation velocity of $v_{\mathrm{coax}}/c = 0.54 \pm 0.10$, broadly consistent with RG-58 specifications. The SDR operates in a linear regime with 29 dB of clipping headroom (RMSE = 0.098 dB). Y-factor calibration with a 310 K hot load and 30 K cold sky yielded receiver noise temperatures $T_{\mathrm{rx}} = 79 \pm 4\,\mathrm{K}$ at LO 1420 MHz and $96 \pm 4\,\mathrm{K}$ at LO 1421 MHz, dominated by hot/cold load temperature uncertainty. Dual-LO ($f_{\mathrm{LO}} = 1420$, 1421 MHz) frequency-switched observations of the Galactic-plane field $(l, b) = (120°, 0°)$ reveal a rich, multi-component HI spectrum spanning -110 to $+20\,\mathrm{km\,s^{-1}}$ in LSR velocity, interpreted as multiple spiral-arm crossings in the Perseus arm region. A Cygnus-X cross-validation field confirms a dominant feature near $-16\,\mathrm{km\,s^{-1}}$ attributed to Local Arm material.

Preface

I would like to apologize for this late work but I had a lot of fun doing it and I hope you enjoy reading it as much as I enjoyed taking data. It devolved into a passion project of sorts and I took way too long in writing it. I am sorry for giving empty promises but I will do my best to juggle this and AY-128 for the next two labs.

1. Introduction

The 21-cm hyperfine line of neutral hydrogen (HI) at $\nu_0 = 1420.40575\,\mathrm{MHz}$ is the most widely used spectral tracer of the interstellar medium (ISM), first detected by Ewen & Purcell (1951). Because the Milky Way's disk is optically thin to 21-cm radiation, the line traces the full column of HI along any line of sight and encodes the radial-velocity distribution of ISM clouds through the Doppler shift. The Big Horn antenna at Campbell Hall, UC Berkeley, provides a simple but complete radio-astronomy signal chain for observing this line with a small horn and low-cost software-defined radio (RTL-SDR).

This report documents a three-stage workflow, following Parsons (2024a): **(1)** *equipment calibration* characterizing coax attenuation, propagation velocity, SDR linearity, and the FIR passband response; **(2)** *temperature calibration* measuring the receiver noise temperature via the Y-factor method with hot and cold load references; and **(3)** *HI observations* frequency-switched line extraction, absolute intensity calibration, Doppler/LSR correction, and Gaussian decomposition.

2. Theoretical Background

2.1. HI 21-cm Hyperfine Transition

The ground state of hydrogen is split by the interaction between the proton and electron magnetic moments into two hyperfine levels: the parallel-spin triplet state ($F = 1$, higher energy) and the antiparallel-spin singlet state ($F = 0$, lower energy). The energy splitting corresponds to a photon frequency of

$$\nu_0 = 1\,420\,405\,751.768\ \mathrm{Hz} \approx 1420.41\ \mathrm{MHz}. \tag{1}$$

The spontaneous emission (21-cm) Einstein A-coefficient is $A_{10} = 2.869 \times 10^{-15}\,\mathrm{s^{-1}}$, corresponding to a radiative lifetime of $\sim 11\,\mathrm{Myr}$. Despite the low transition probability, the abundance of HI ($\sim 10^{21}\,\mathrm{cm^{-2}}$ in the Galactic plane) makes the line bright. The line was first detected by Ewen & Purcell 1951 and has since become the premier tracer of Galactic HI structure (Dickey & Lockman 1990).

In the optically thin regime, the observed line brightness temperature is

$$T_B(\nu) = T_s \left[1 - e^{-\tau(\nu)}\right] \approx T_s\,\tau(\nu), \tag{2}$$

with spin temperature T_s and optical depth $\tau(\nu)$.

2.2. Thermal (Doppler) Line Broadening

Thermal motion of hydrogen atoms at temperature T broadens the line via the Doppler effect. The resulting profile is Gaussian with

$$\sigma_v = \sqrt{\frac{k_B T}{m_H}}, \qquad \mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma_v, \tag{3}$$

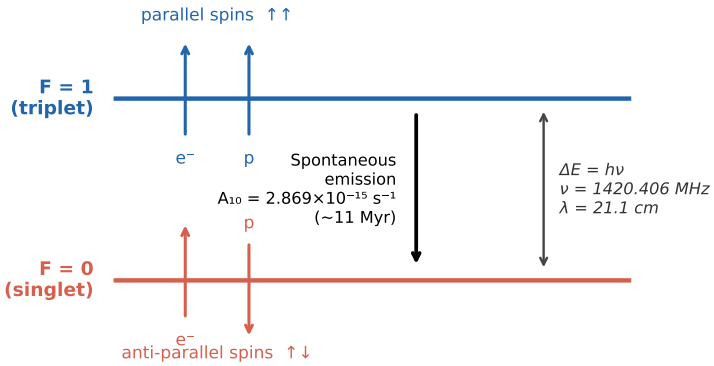

Fig. 1.— Energy-level diagram of the hydrogen 21-cm hyperfine transition. The triplet ($F = 1$) and singlet ($F = 0$) states are split by $\Delta E = h\nu_0 \approx 5.87\,\mu$eV. Spontaneous emission produces the 21-cm photon at $\nu_0 = 1420.41$ MHz.

where $m_H = 1.673 \times 10^{-27}$ kg. At $T = 100$ K, FWHM $\approx 2.1\,\mathrm{km\,s^{-1}}$; at $T = 8000$ K, FWHM $\approx 19\,\mathrm{km\,s^{-1}}$. Turbulent broadening adds in quadrature and dominates for many ISM components.

2.3. Receiver Noise and System Temperature

A radiometric channel records power proportional to the system temperature:

$$P_j = G_j\, T_{\mathrm{sys}}(\nu_j), \qquad T_{\mathrm{sys}} = T_{\mathrm{rx}} + T_{\mathrm{sky}}, \tag{4}$$

where G_j is the total chain gain and T_{rx} is the (frequency-dependent) receiver contribution. For a single-channel noise measurement, the radiometer equation gives the thermal fluctuation Wilson et al. 2013

$$\delta T_{\mathrm{chan}} = \frac{T_{\mathrm{sys}}}{\sqrt{\Delta\nu\,\tau}}, \tag{5}$$

with channel bandwidth $\Delta\nu$ and integration time τ. For this system: $\Delta\nu = 78.125$ Hz (SDR, 32768-bin grid), $\Delta\nu_{\mathrm{cal}} = 312.5$ Hz (8192-bin grid), $\tau = 419.43$ s (combined-spectrum integration).

2.4. Y-Factor Calibration

The Y-factor method measures T_{rx} by comparing power outputs when the receiver is fed with two known-temperature sources (Friis 1944):

$$Y = \frac{P_{\mathrm{hot}}}{P_{\mathrm{cold}}} = \frac{T_{\mathrm{hot}} + T_{\mathrm{rx}}}{T_{\mathrm{cold}} + T_{\mathrm{rx}}}, \tag{6}$$

solving for

$$T_{\mathrm{rx}} = \frac{T_{\mathrm{hot}} - Y\,T_{\mathrm{cold}}}{Y - 1}. \tag{7}$$

Error propagation through Equation (7) gives

$$\sigma^2_{T_{\mathrm{rx}}} = \left(\frac{\partial T_{\mathrm{rx}}}{\partial T_{\mathrm{hot}}}\right)^2 \sigma^2_{T_h} + \left(\frac{\partial T_{\mathrm{rx}}}{\partial T_{\mathrm{cold}}}\right)^2 \sigma^2_{T_c} + \left(\frac{\partial T_{\mathrm{rx}}}{\partial Y}\right)^2 \sigma^2_Y. \tag{8}$$

The three contributions are measurement, load-temperature, and hardware-systematic terms, which are decomposed individually in Section 5.3.

2.5. Frequency Switching and Bandpass Removal

Frequency switching removes the slowly varying receiver bandpass $G_j(\nu)$ by dividing on-source and off-source spectra taken at two LO settings (Parsons 2024b):

$$R(\nu_k) = \frac{P_{1420}(\nu_k)}{P_{1421}(\nu_k)}, \qquad y_R = R - 1. \tag{9}$$

Since the bandpass shape is approximately the same at both LO settings (same receiver chain), the ratio R cancels G_j and isolates the spectral line relative to the continuum. The inverse-ratio variant

$$y_{\mathrm{inv}} = \frac{1}{R} - 1 = \frac{P_{1421}}{P_{1420}} - 1 \tag{10}$$

recovers the same line information with a sign flip. Absolute intensity scale is restored by multiplying y_R by the system temperature reference computed from the cold-sky spectrum:

$$T_{\mathrm{line}}(\nu_k) = y_R(\nu_k)\, T_{\mathrm{sys,ref}}(\nu_k), \tag{11}$$

$$T_{\mathrm{sys,ref}}(\nu_k) = P_{\mathrm{obs}}(\nu_k)\frac{T_{\mathrm{cold}} + T_{\mathrm{rx}}}{P_{\mathrm{cold,ref}}(\nu_k)}. \tag{12}$$

2.6. Transmission-Line Theory

A coaxial cable is a two-conductor transmission line with characteristic impedance $Z_0 = 50\,\Omega$ (Parsons 2024c). When a traveling wave reaches a load impedance $Z_L \neq Z_0$, a reflection is generated with voltage reflection coefficient

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}. \tag{13}$$

For an open circuit ($Z_L \to \infty$): $\Gamma = +1$ (full reflection, same polarity); for a short circuit ($Z_L = 0$): $\Gamma = -1$ (full reflection, inverted); for a matched load ($Z_L = Z_0$): $\Gamma = 0$ (no reflection). The round-trip delay for a cable of length L and propagation velocity v_p:

$$\Delta t = \frac{2L}{v_p}, \qquad v_p = \frac{2L}{\Delta t}. \tag{14}$$

Cable attenuation is modeled as a linear function of length L at fixed frequency:

$$P_{\mathrm{rx}}(L) = B - \alpha L \quad [\mathrm{dB}], \tag{15}$$

where α is the attenuation coefficient (dB/m) and B is an intercept. An unknown-length cable can be inferred from a measured power loss:

$$L_{\mathrm{unknown}} = \frac{B - P_{\mathrm{obs}}}{\alpha}. \tag{16}$$

2.7. Coordinate Frames and Pointing

Target positions in Galactic coordinates (l, b) are converted to equatorial J2000 (α, δ) using the IAU 1958 Galactic convention of Blaauw et al. (1960), carried to the FK5/J2000 equator and equinox following Murray (1989):

$$\hat{n}_{\rm eq} = M_{\rm G\to J2000}\, \hat{n}_{\rm gal}, \tag{17}$$

where $\hat{n}_{\rm gal} = (\cos b \cos l,\ \cos b \sin l,\ \sin b)^{\mathsf{T}}$ and $M_{\rm G\to J2000}$,

$$\begin{bmatrix} -0.05487554 & 0.49410945 & -0.86766614 \\ -0.87343710 & -0.44482959 & -0.19807639 \\ -0.48383499 & 0.74698225 & 0.45598379 \end{bmatrix}. \tag{18}$$

The corresponding FK5/J2000 pole parameters are $(\alpha_{\rm NGP}, \delta_{\rm NGP}) = (192.85948^\circ, 27.12825^\circ)$ and $l_\Omega = 32.93192^\circ$ (equivalently $l_{\rm NCP} = 122.93192^\circ$).

Given Local Sidereal Time (LST) at longitude λ, the hour angle is $H = {\rm LST} - \alpha$, and the topocentric altitude and azimuth are

$$\sin({\rm Alt}) = \sin\delta \sin\phi + \cos\delta \cos\phi \cos H, \tag{19}$$

$$\mathrm{Az} = \arctan\left(\frac{-\cos\delta \sin H}{\sin\delta \cos\phi - \cos\delta \cos H \sin\phi}\right), \tag{20}$$

where $\phi = 37.873^\circ$N is the observatory latitude. The time-standard pipeline is

$$t_{\rm unix} \xrightarrow{\rm JD} {\rm JD} \xrightarrow{\rm GMST} {\rm GMST} + \lambda \to {\rm LST}. \tag{21}$$

Implementations use `ugradio.timing` for JD/LST conversion.

2.8. Doppler Effect and LSR Frame

The radio-definition Doppler velocity is

$$v_{\rm radio} = c\,\frac{\nu_0 - \nu}{\nu_0}. \tag{22}$$

Observed velocities are topocentric; to convert to the Local Standard of Rest (LSR), we subtract the projection of the Sun's peculiar velocity onto the line of sight:

$$v_{\rm LSR} = v_{\rm topo} + \Delta v_{\rm LSR}, \tag{23}$$

where $\Delta v_{\rm LSR}$ is computed from the IAU-standard solar apex ($l = 56^\circ$, $b = 23^\circ$, $v = 20\,{\rm km\,s^{-1}}$) and the pointing metadata via `ugradio.doppler.get_projected_velocity`.

3. Instrument Description

The Big Horn receiver chain (Figure 3) consists of three stages: a rooftop RF front-end, a coaxial transmission line, and a laboratory back-end with SDR digitization.

The horn antenna has aperture dimensions estimated to be roughly $0.80\,{\rm m} \times 0.90\,{\rm m}$ (area $= 0.72\,{\rm m}^2$). Following the lab handout's rule-of-thumb for a reasonably well-matched horn (Parsons 2024a), we adopt a provisional aperture efficiency $\eta_{\rm ap} \approx 0.6$–$0.7$; this is an assumed prior rather than a direct measurement. The RTL-SDR (RTL2832U chipset) digitizes at 2.56 MS/s, producing either an 8192-channel FFT grid at $\Delta\nu = 312.5\,{\rm Hz}$ resolution or a 32768-channel grid at $\Delta\nu = 78.125\,{\rm Hz}$ over the same 2.56 MHz bandwidth. The per-observation integration time is 104.86 s. The RTL2832U applies a 32-tap symmetric FIR filter as part of its internal decimation chain to suppress aliasing before FFT channelization; combined with the firmware's channel-summing operation, this produces a frequency-dependent passband that rolls off toward the band edges. As such, for line analysis we focus on the flatter central region within roughly ± 0.8 MHz of the LO.

A Mini-Circuits ZFSC-2-372-S+ two-way splitter in the bench configuration introduces Port 1 loss of 3.374 dB and Port 2 loss of 3.348 dB at 1420 MHz. The SDR internal attenuator was set to 3.0 dB during observations.

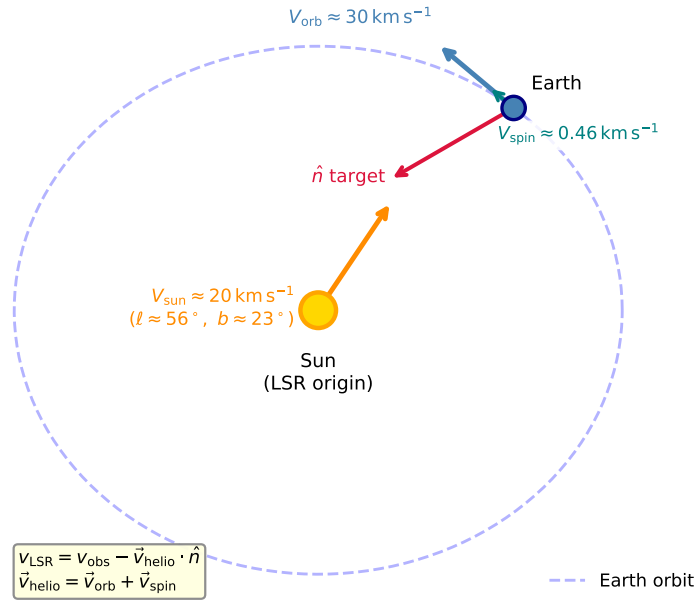

Fig. 2.— Geometry of the LSR velocity correction. The Sun's motion relative to the LSR projects onto the line of sight to the target, adding a systematic offset that depends on the pointing direction and observation epoch.

4. Equipment Calibration

4.1. Observation Geometry and Pointing

The primary science target is $(l, b) = (120^\circ, 0^\circ)$. Applying the Galactic→J2000 rotation matrix from Section 2.7 gives equatorial coordinates $(\alpha, \delta) = (96.45^\circ, 62.73^\circ)$ in J2000. For the observation epoch (JD = 2461105.386, LST = 0.0072 h), the topocentric coordinates are Alt = 64.68°, Az = 7.03°, confirming that the target is well above the horizon and near transit.

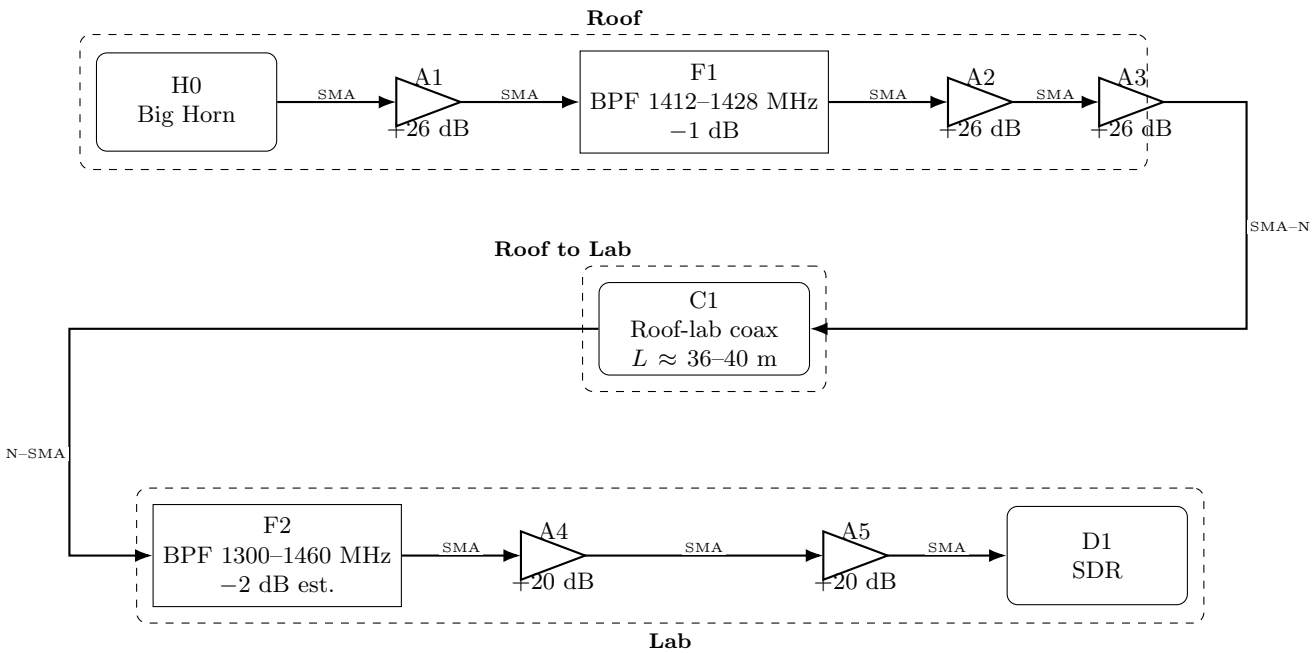

Fig. 3.— RF signal chain from the Big Horn antenna to the RTL-SDR. Rooftop section: Big Horn → A1/A2/A3 (Mini-Circuits ZKL-33ULN-S+, ≈ +26 dB each) → F1 (Reactel 1420-15S11, passband 1412–1428 MHz, ≈ −1 dB). Transmission line: roof–lab coaxial cable, $L \approx 36$–40 m (Section 4.2). Lab section: F2 (K&L Microwave, passband 1300–1460 MHz, ≈ −2 dB) → A4/A5 (lab amplifiers, ≈ +20 dB each) → RTL-SDR. Net bench gain Horn→SDR: −6.0 dB.

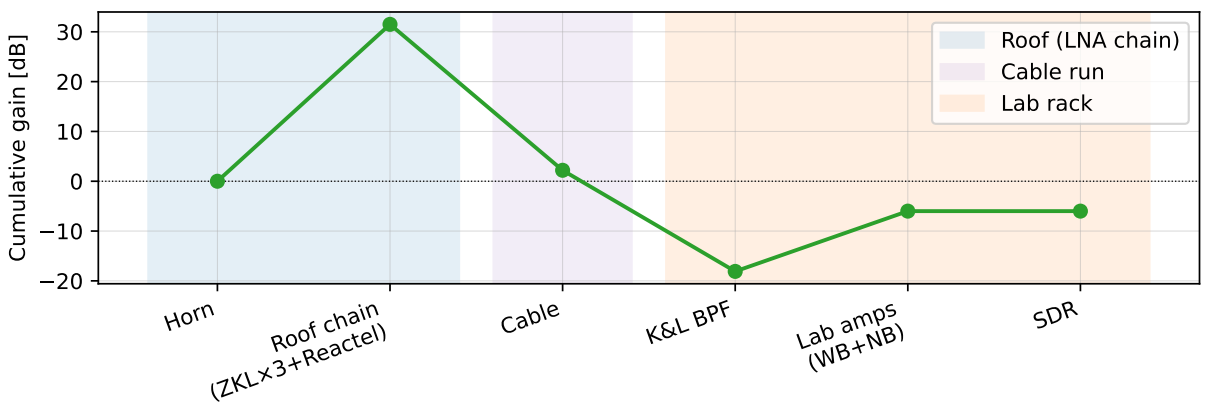

Fig. 4.— Cumulative signal chain gain/loss from horn to SDR, showing contributions of each component. The net gain measured at the bench (Horn→SDR, 2026-03-07) is −6.0 dB, reflecting the combined effect of three rooftop amplifiers, two filters, two lab amplifiers, and approximately 36–40 m of roof–lab coax.

Time-standard conversions used throughout:

$$\mathrm{PST} = \mathrm{UTC} - 8\ \mathrm{hr}, \tag{24}$$

$$\mathrm{JD} = \texttt{ugradio.timing.julian_date}(t_{\mathrm{unix}}), \tag{25}$$

$$\mathrm{LST} = \texttt{ugradio.timing.lst}(\mathrm{JD},\ \lambda_{\mathrm{obs}}), \tag{26}$$

with observatory coordinates $(\phi_{\mathrm{obs}}, \lambda_{\mathrm{obs}}, h_{\mathrm{obs}}) = (37.873°\mathrm{N}, -122.257°, \approx 101\ \mathrm{m})$ above sea level. The pointing was validated by comparing `ugradio` outputs against `astropy` for the same JD; agreement was within numerical precision.

4.2. Cable Attenuation Characterization

4.2.1. Experimental Setup

To characterize the coaxial cable attenuation coefficient α (dB/m), power measurements were made through known-length RG-58 cable segments at both LO frequencies (1420 and 1421 MHz) (bench layout: Appendix 16). A shared-slope linear model was fit jointly to both LO datasets:

$$P_{\mathrm{rx},i}(L) = B_i - \alpha L, \quad i \in \{1420, 1421\}, \tag{27}$$

where B_i is a frequency-dependent intercept and α is shared. One outlier cable run was removed during quality screening (8 of 9 runs retained based on residual diagnostics).

4.2.2. Results: SDR Method

The SDR-path fit gives:

$$\alpha_{\mathrm{SDR}} = 0.73 \pm 0.13 \text{ dB/m}, \tag{28}$$

with a 95% confidence interval of $[0.47, 0.99]$ dB/m.

4.2.3. Cross-Check: Power-Meter Method

An independent power-meter measurement of the same cable set yields:

$$\alpha_{\mathrm{meter}} = 0.80 \pm 0.03 \text{ dB/m}, \tag{29}$$

with a 95% confidence interval of $[0.74, 0.86]$ dB/m. The two estimates differ by 0.07 dB/m (0.6σ), consistent within uncertainties. The power-meter value is substantially more precise due to its direct, analog calibrated measurement versus the SDR's indirect spectral ratio.

4.2.4. Unknown Roof–Lab Cable Length

Using the fitted SDR intercept, the unknown roof–lab cable length is inferred as (Appendix 17):

$$L_{\mathrm{SDR}} = 40 \pm 7 \text{ m}. \tag{30}$$

Independent estimates are: $L_{\mathrm{meter}} = 36 \pm 1$ m (power-meter intercept) and $L_{\mathrm{bench}} = 36 \pm 1$ m (bench analog measurement). The bench and meter estimates are in excellent agreement, while the SDR estimate is $\sim 0.55\sigma$ higher. We adopt $L \approx 36$–40 m as a conservative range, with $L = 36$ m as the best point estimate from the more precise analog measurements.

The total attenuation contribution of the 36 m roof–lab cable at 1420 MHz is approximately $36 \times 0.80 \approx 29$ dB, which is partially compensated by the three rooftop and two lab amplifiers.

4.3. Termination States and Coax Propagation Velocity

4.3.1. Reflectometry Measurement

A square-wave generator was connected to one end of a coaxial cable of known length $L = 40$ m (primary reflectometry branch; bench layout: Appendix 18). The reflected echo is observed with a modular edge separation

$$\Delta t_{\mathrm{mod}} = 98.0 \text{ ns}. \tag{31}$$

Because the drive waveform is a 2.5 MHz square wave with period $T = 400.0$ ns, the physical round-trip delay is branch-resolved as

$$\Delta t_{\text{round-trip}} = \Delta t_{\mathrm{mod}} + nT = 498.0 \text{ ns}, \tag{32}$$

where the $n = 1$ branch is the smallest solution consistent with a coax-like propagation speed for a 40 m cable. The coax propagation velocity is therefore

$$v_{\mathrm{coax}} = \frac{2L}{\Delta t} = 1.609 \times 10^8 \text{ m/s}, \tag{33}$$

corresponding to a velocity factor

$$v_{\mathrm{coax}}/c = 0.54 \pm 0.10. \tag{34}$$

For the Pasternack RG58C/U cable used here, the nominal velocity factor is $v/c = 0.659$ (Pasternack 2026); our measurement is $\sim 1.2\sigma$ below this, but still consistent within uncertainty.

4.3.2. Termination States

The cable was tested in four states to characterize impedance behavior: *open end* ($\Gamma = +1$, full positive reflection), *50 Ω matched termination* ($\Gamma \approx 0$, minimal reflection), *SDR-connected unpowered*, and *SDR-connected powered*. The terminated state showed the expected near-zero reflection, confirming proper 50 Ω source impedance. The SDR input impedance in both powered and unpowered states showed small but non-negligible reflections, indicating slight impedance mismatch at 1420 MHz even though the SMA connector on the SDR is designed for 50 Ω impedance.

4.4. SDR Characterization

4.4.1. Gain Linearity

The RTL-SDR gain response was swept over a range of input power levels to characterize the linear operating regime (bench layout: Appendix 19). The key results are:

- Highest unclipped gain setpoint: 13.0 dBm
- First clipped setpoint: 15.5 dBm
- Clipping headroom: 29.0 dB (from the observed sky power level)
- Linearity RMSE in linear regime: 0.1 dB

As such, sky observations should operate well within a linear gain response.

4.4.2. FIR Passband Response

The RTL2832U digitizer includes a 32-tap symmetric FIR decimation filter. Rather than measuring this response with a swept tone, we constructed a synthetic hardware-response diagnostic from the published FIR coefficients, then multiplied by a symmetric model of the undocumented summing stage and applied the combined response to a noise input on the output-frequency grid. Figure 9 compares two summing coefficients: Aaron Parsons's guessed coefficients and a regression-optimized set fit to flatten the observed passband. Key diagnostic results:

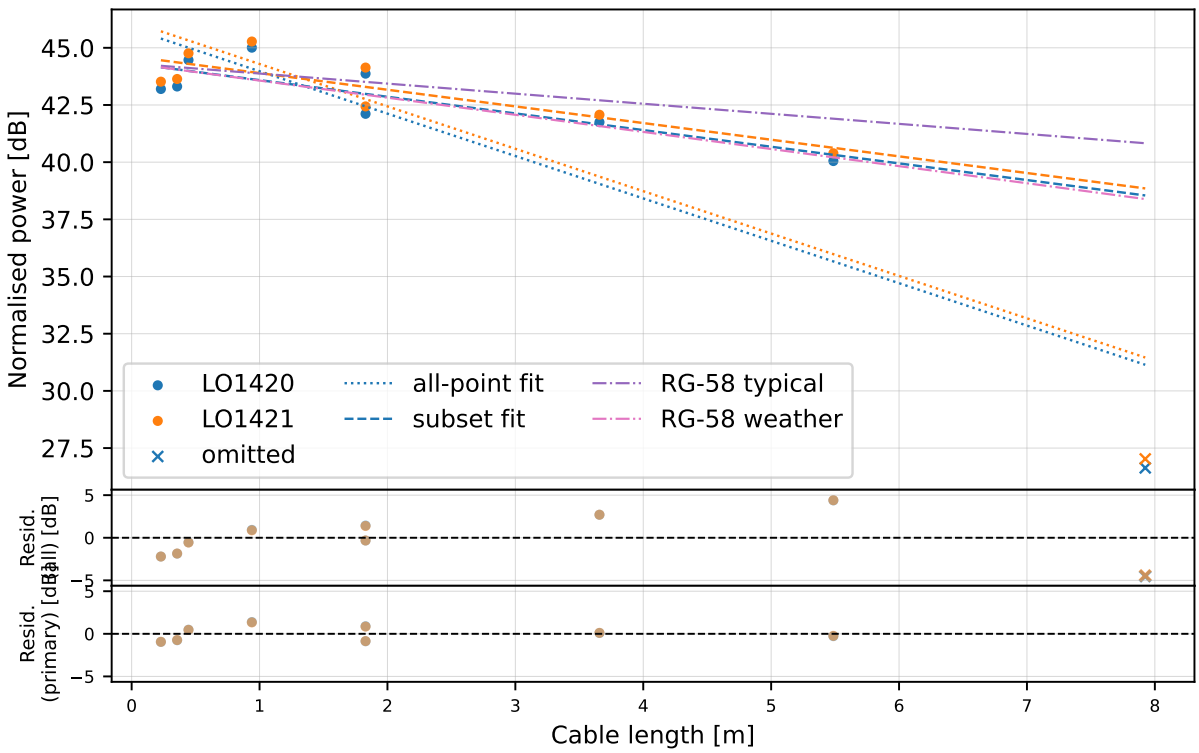

Fig. 5.— Normalized SDR power versus cable length for the LO 1420 and LO 1421 bench measurements. The dotted lines show fits using all cable runs, while the dashed lines show the adopted shared-slope fit after omitting the 26-ft cable outlier marked with ×. Dash-dotted lines give representative RG-58 reference attenuation trends for comparison. The lower two panels show the corresponding residuals for the two fit choices. Notably, the fit omitting the strong outlier matches well with an RG-58 weather-resistant reference attenuation.

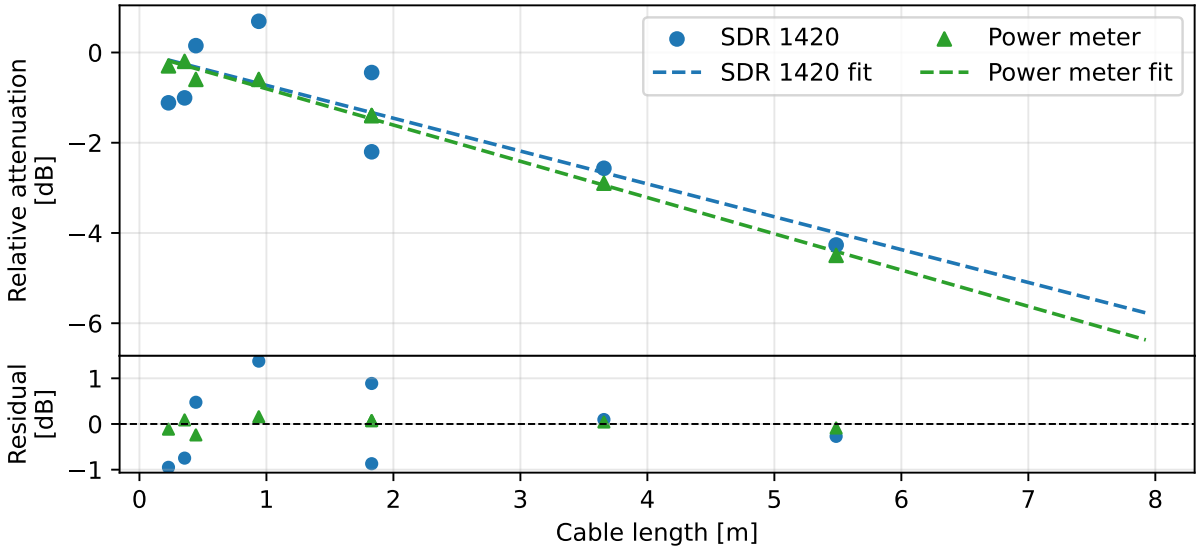

Fig. 6.— Cable attenuation cross-check using a calibrated power meter. The fitted slope $\alpha_{\rm meter} = 0.80 \pm 0.03\,{\rm dB/m}$ is statistically consistent with the SDR estimate within $\sim 0.6\sigma$.

- FIR-only edge/center response ratio: 0.237 (i.e., edge channels have $\sim 6.2\,{\rm dB}$ lower gain)
- Science analysis should be safe to focus on the flatter central region within $\pm 0.8\,{\rm MHz}$ of the LO

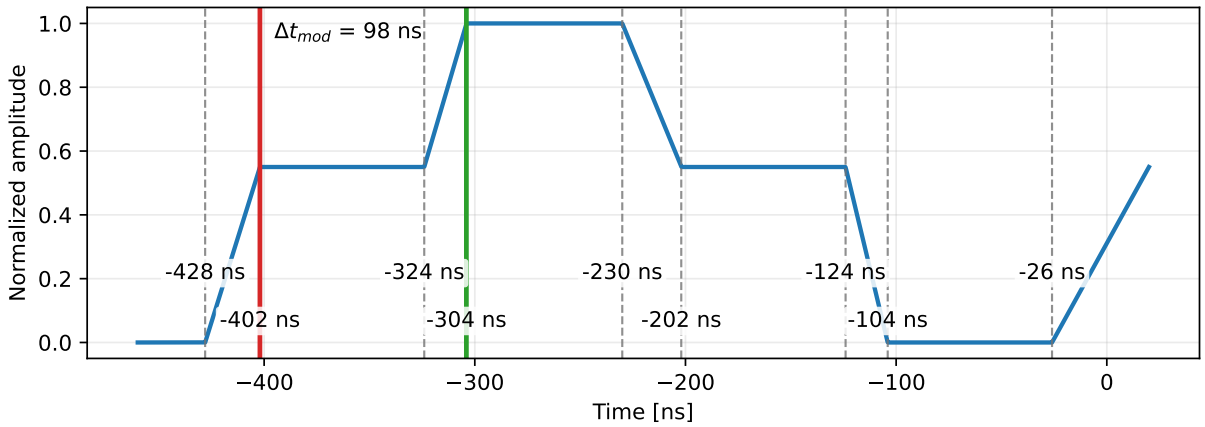

Fig. 7.— Square-wave reflectometry trace showing the incident signal and reflected pulse from the open cable end ($\Gamma = +1$). The measured modular edge separation is $\Delta t_{\rm mod} = 98.0\,$ns; with the 2.5 MHz square-wave period $T = 400.0\,$ns and branch $n = 1$, this gives the physical round-trip delay $\Delta t = 498.0\,$ns and $v_{\rm coax}/c = 0.54 \pm 0.10$.

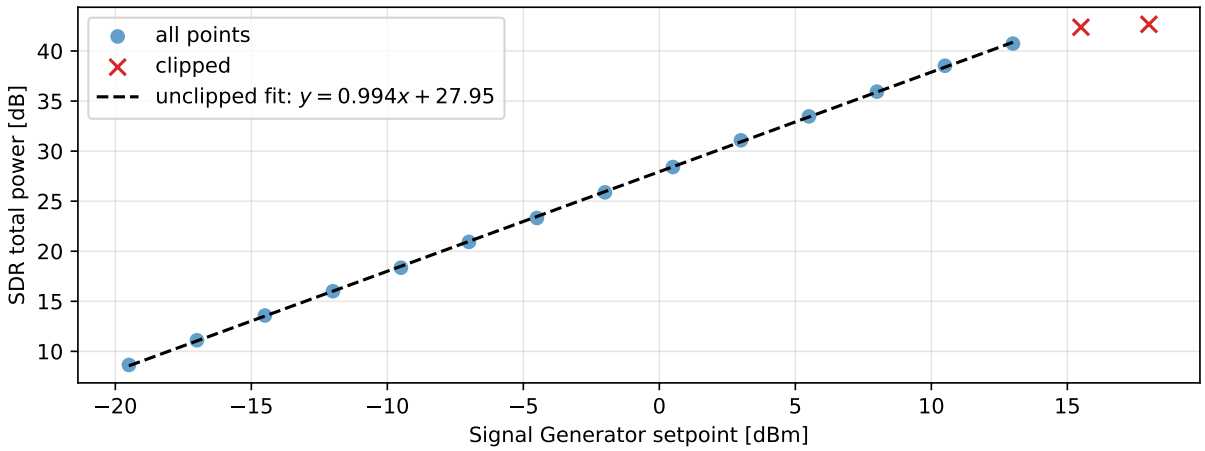

Fig. 8.— SDR power response as a function of input setpoint. The linear regime (solid line) transitions to saturation/clipping above 13.0 dBm. Sky observations at the current gain setting have 29 dB of headroom below the clipping threshold.

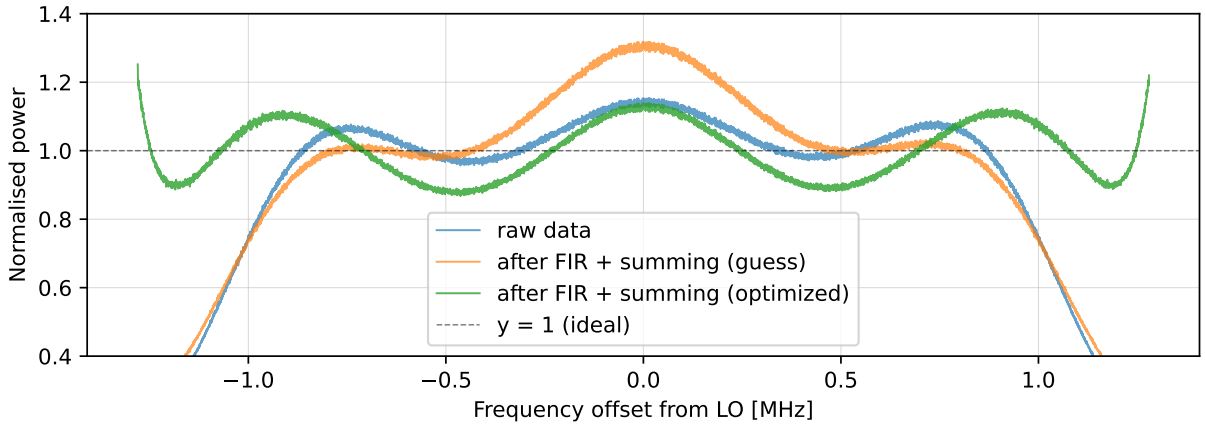

Fig. 9.— Synthetic FIR+summing diagnostic across the RTL-SDR passband. The blue trace is a normalized noise-input realization, the orange curve applies the published FIR coefficients plus Aaron Parsons's guessed summing coefficients, and the green curve uses regression-optimized summing coefficients on top of the same FIR model.

5. Temperature Calibration

5.1. Calibration Procedure

The Y-factor method (Friis 1944) was used to measure the receiver noise temperature using two calibration loads (setup: Appendix 20):

- **Hot load:** A human body (hand blocking the horn aperture) at $T_{\rm hot} = 310 \pm 5\,{\rm K}$.
- **Cold load:** Cold sky pointed toward a low HI column region ($l \approx 165°$, $b \approx 36°$) assumed at $T_{\rm cold} = 30 \pm 2\,{\rm K}$.

The cold reference was observed just before hot measurements to minimize gain drift or sky variability due to weather. Per-channel $T_{\rm rx}(\nu)$ profiles were computed by applying Equation (7) to each spectral channel independently, then averaged to obtain the broadband $T_{\rm rx}$ values.

5.2. RFI Screening

Before computing Y-factors, both hot and cold spectra were screened for radio-frequency interference (RFI) using an iterative sigma-clipping procedure with $n_\sigma = 5$. The resulting common mask retains 8186 channels (LO 1420) and 8187 channels (LO 1421) out of 8192 total — more than 99.9% of channels pass the screening.

5.3. Y-Factor Results

Table 1 summarizes the Y-factor measurement results.

Table 1: Y-factor calibration results at both LO settings. The σ column is decomposed into two contributions: hot/cold load temperatures (load), and hardware-response systematic (hw).

LO	Y	$T_{\rm rx}$ (K)	$\sigma_{\rm load}$ (K)	$\sigma_{\rm hw}$ (K)
1420 MHz	3.56	79 ± 4	3	1
1421 MHz	3.22	96 ± 4	3	1

The LO 1421 receiver temperature is 17 K higher than LO 1420 (96 vs. 79 K). This could be due to gain variations of the receiver chain with frequency, as the bandpass response differs slightly at the two LO settings.

5.4. System Temperature and Thermal Noise

The total system temperature includes sky and receiver contributions:

$$T_{\rm sys}(1420) \approx T_{\rm rx} + T_{\rm sky} \approx 79 + 30 = 109\ {\rm K}, \tag{35}$$

$$T_{\rm sys}(1421) \approx 96 + 30 = 126\ {\rm K}, \tag{36}$$

where $T_{\rm sky} \approx 30\,{\rm K}$ is the adopted sky-temperature estimate for the diffuse background at 1420 MHz in the standard field direction.

The per-channel thermal noise (radiometer equation) for the combined spectrum ($\tau = 419.43\,{\rm s}$, $\Delta\nu = 78.125\,{\rm Hz}$ for 32768-channel spectra; equivalent to $\Delta\nu = 312.5\,{\rm Hz}$ for the 8192-channel calibration grid) is:

$$\delta T_{\rm chan}(1420) = 603.2 \pm 19.4\ {\rm mK}, \tag{37}$$

$$\delta T_{\rm chan}(1421) = 697.6 \pm 21.2\ {\rm mK}. \tag{38}$$

6. HI 21-cm Line Observations and Analysis

6.1. Observation Overview

Two sky fields were observed (Parsons 2024a):

1. **Standard field**: $(l, b) = (120°, 0°)$ — primary science target in the Galactic plane, Perseus arm direction. Observed on 2026-03-07, JD = 2461105.386, LST = 0.007 h, Alt = 64.68°, Az = 7.03°.
2. **Cygnus-X field**: observed for cross-validation, at the SIMBAD-resolved Cygnus X position (Wenger et al. 2000) $(l, b) = (79.50°, 1.00°)$, providing an independent sight line through the Galactic plane. Observed 2026-03-07, JD = 2461105.400, LST = 0.092 h.

Both fields were observed with dual-LO (1420 and 1421 MHz) frequency switching.

6.2. Frequency Switching and Ratio Profiles

Following the theory in Section 2.5 (Parsons 2024b), the frequency-switched ratio $R(\nu_k) = P_{1420}(\nu_k)/P_{1421}(\nu_k)$ removes the slowly varying receiver bandpass. The ratio profiles $y_R = R - 1$ and $y_{\rm inv} = 1/R - 1$ were computed channel by channel after RFI masking, then smoothed with a 31-channel running mean ($\approx 0.51\,{\rm km/s}$ kernel). A representative smoothing comparison is shown in Figure 12, where 9-channel sliding mean and median filters are applied to a normalized LO 1420 PSD. The two estimators track each other closely across the usable band; the median is slightly more resistant to small fluctuations, while the mean follows narrow spectral structure somewhat more closely. Because explicit RFI masking is applied before line extraction, the final science profiles use the running mean.

LSR corrections were applied independently to each spectrum using `ugradio.doppler` with the pointing direction and observation epoch:

$$\Delta v_{\rm LSR}(1420) = -14.12\ {\rm km\,s^{-1}}. \tag{39}$$

6.3. Intensity Calibration and Temperature Profile

The calibrated line temperature is computed via Path B (temperature-domain calibration):

$$T_{\rm line}(\nu_k) = (R - 1)\, T_{\rm sys,ref}(\nu_k), \tag{40}$$

where $T_{\rm sys,ref}(\nu_k)$ is determined from the cold-reference spectrum and the measured $T_{\rm rx}$. The synthetic FIR+summing hardware response is divided out channel by channel.

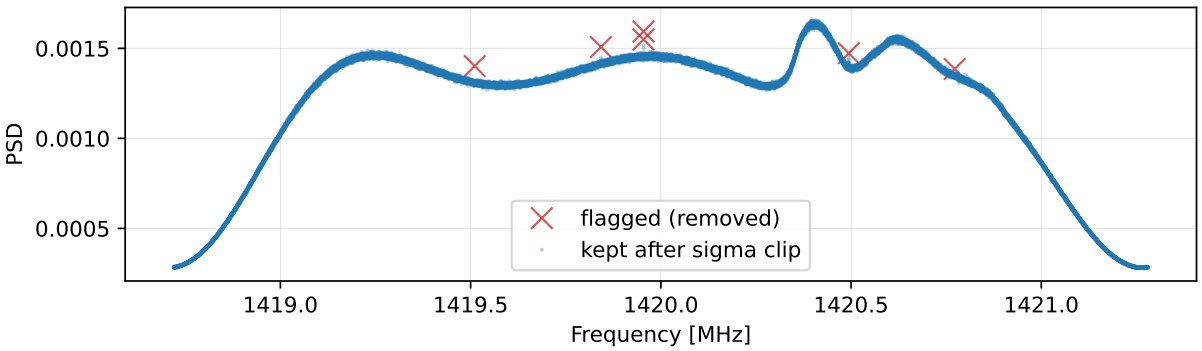

Fig. 10.— RFI sigma-clipping mask for the temperature calibration spectra. Blue channels are retained; masked channels (red) correspond to RFI spikes and hardware response edges. More than 99.9% of channels pass the combined $n_\sigma = 5$ and hardware-response-support cuts.

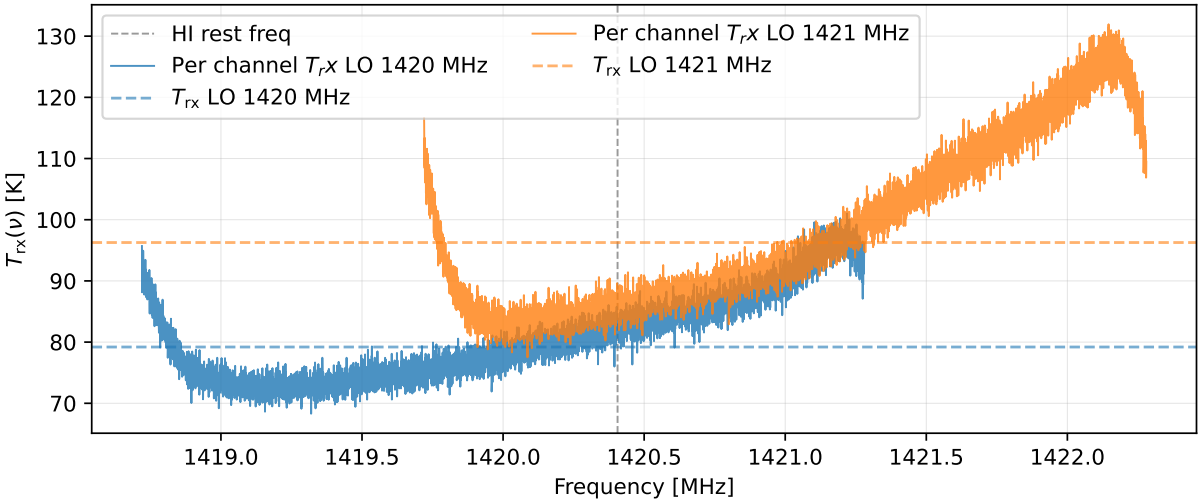

Fig. 11.— Per-channel receiver noise temperature $T_{\rm rx}(\nu)$ for both LO settings, computed channel-by-channel from the Y-factor. The frequency dependence indicates non-uniform gain across the passband, particularly near the band edges where the FIR response drops.

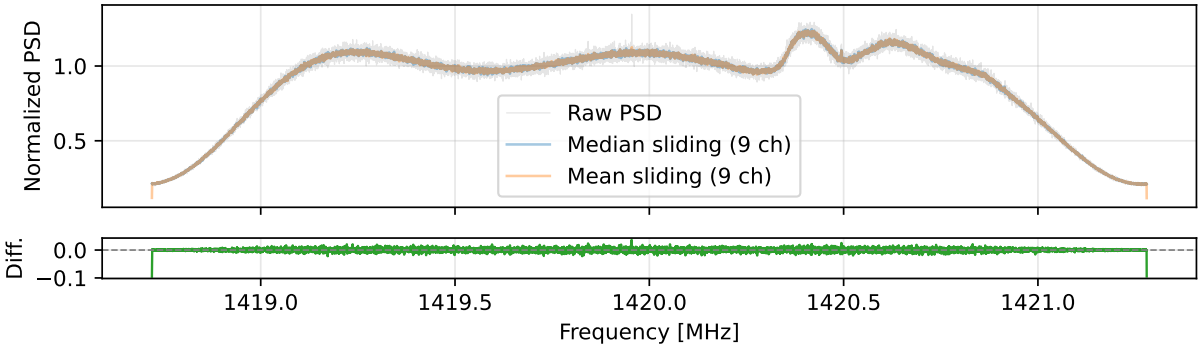

Fig. 12.— Comparison of sliding mean and sliding median smoothing on a representative normalized LO 1420 PSD. The top panel shows the raw PSD together with 9-channel sliding-median and sliding-mean estimates; the lower panel shows their difference, Mean−Median. Over the central passband the two smoothed curves are nearly indistinguishable, with only noticeable deviations only near isolated excursions and edge channels.

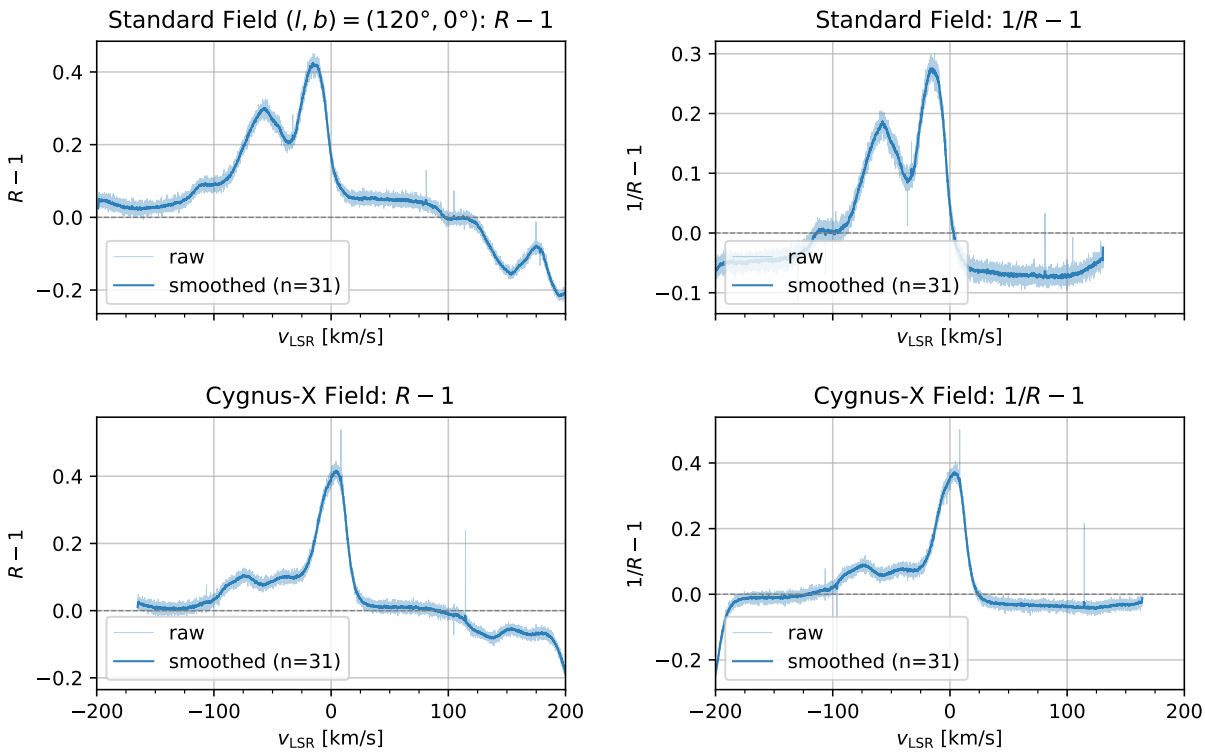

Fig. 13.— Frequency-switched ratio profiles for both fields. Top row: standard field $(l, b) = (120°, 0°)$. Bottom row: Cygnus-X field. Left column: $R-1 = P_{1420}/P_{1421}-1$ plotted against LO 1420 LSR velocity. Right column: $1/R-1 = P_{1421}/P_{1420}-1$ plotted against LO 1421 LSR velocity. Both panels show the raw profile (pale) and 31-channel smoothed profile (solid). The dominant HI emission feature near -10 to $+20\,\mathrm{km\,s^{-1}}$ LSR is clearly detected in both fields.

6.4. Joint Baseline and Gaussian Decomposition

A multi-component Gaussian models were fit jointly with a low-order polynomial baseline to both the R-branch (y_R profile) and the $R_{\rm inv}$ branch ($y_{\rm inv}$ profile) for each field. The model is

$$y(v) = \sum_{k=1}^{N} A_k \exp\left[-\frac{(v-\mu_k)^2}{2\sigma_k^2}\right] + \sum_{m=0}^{M} c_m\,(v/100)^m, \tag{41}$$

with FWHM $= 2\sqrt{2\ln 2}\,\sigma_k$, constrained to $3\,\mathrm{km/s} \leq \mathrm{FWHM} \leq 35\,\mathrm{km/s}$. Additionally, to prevent fitting unphysical solutions, component amplitudes were required to remain non-negative, centroids were confined to the adopted velocity fit window, and widths were bounded to the range above. This trades a formally lower-χ^2_ν unconstrained solution for one whose peak amplitudes, centroid velocities, and FWHM values remain physically interpretable as HI components. The model grid included $N \in \{2, 3, 4\}$ components and polynomial orders $M \in \{1, 2\}$. All selected solutions used four Gaussian components. Path A selected $M = 2$ in all four cases, while Path B selected $M = 1$ for the Cygnus-X R branch and $M = 2$ elsewhere. Figures 14 and 15 show the selected joint Gaussian+baseline decompositions for the standard and Cygnus-X fields, respectively.

Tables 2 list the best-fit parameters for the selected four-component joint models.

Monte Carlo perturbations of the selected calibrated-fit solutions provide a model uncertainty estimate. For the standard field, the effective profile centroid and width are comparatively unstable in the R branch ($\sigma_{\rm cent} \approx 5.6\,\mathrm{km\,s^{-1}}$, $\sigma_{\rm FWHM,eff} \approx 5.2\,\mathrm{km\,s^{-1}}$) because the broad negative-velocity components are partially degenerate and two widths sit at the $35\,\mathrm{km\,s^{-1}}$ upper bound. The $R_{\rm inv}$ branch is internally much tighter (0.42 and $0.65\,\mathrm{km\,s^{-1}}$), but its centroid shifts substantially relative to R.

6.5. Cygnus-X Cross-Validation

The Cygnus-X field shows a dominant HI component at $v_0 = -16.2\,\mathrm{km/s}$ LSR (FWHM $= 26.0\,\mathrm{km/s}$, $A = 13.4\,\mathrm{K}$), consistent with the standard field dominant feature at $-16.0\,\mathrm{km/s}$ and is broadly consistent with the negative-velocity Cygnus-X North / DR21 layer resolved in higher-angular-resolution CO surveys (Schneider et al. 2006; Gottschalk et al. 2012). Within the selected R model, Monte Carlo perturbations give effective whole-profile uncertainties of $\sigma_{\rm cent} \approx 0.07\,\mathrm{km\,s^{-1}}$ and $\sigma_{\rm FWHM,eff} \approx 0.11\,\mathrm{km\,s^{-1}}$, while the dominant $-16.2\,\mathrm{km\,s^{-1}}$ component itself has fitted parameter uncertainties of about $\sigma_{v_0} \approx 0.03\,\mathrm{km\,s^{-1}}$ and $\sigma_{\rm FWHM} \approx 0.03\,\mathrm{km\,s^{-1}}$. The alternative $R_{\rm inv}$ Cygnus-X solution is noticeably less stable ($\sigma_{\rm cent} \approx 0.97\,\mathrm{km\,s^{-1}}$, $\sigma_{\rm FWHM,eff} \approx 1.94\,\mathrm{km\,s^{-1}}$). Thus, we

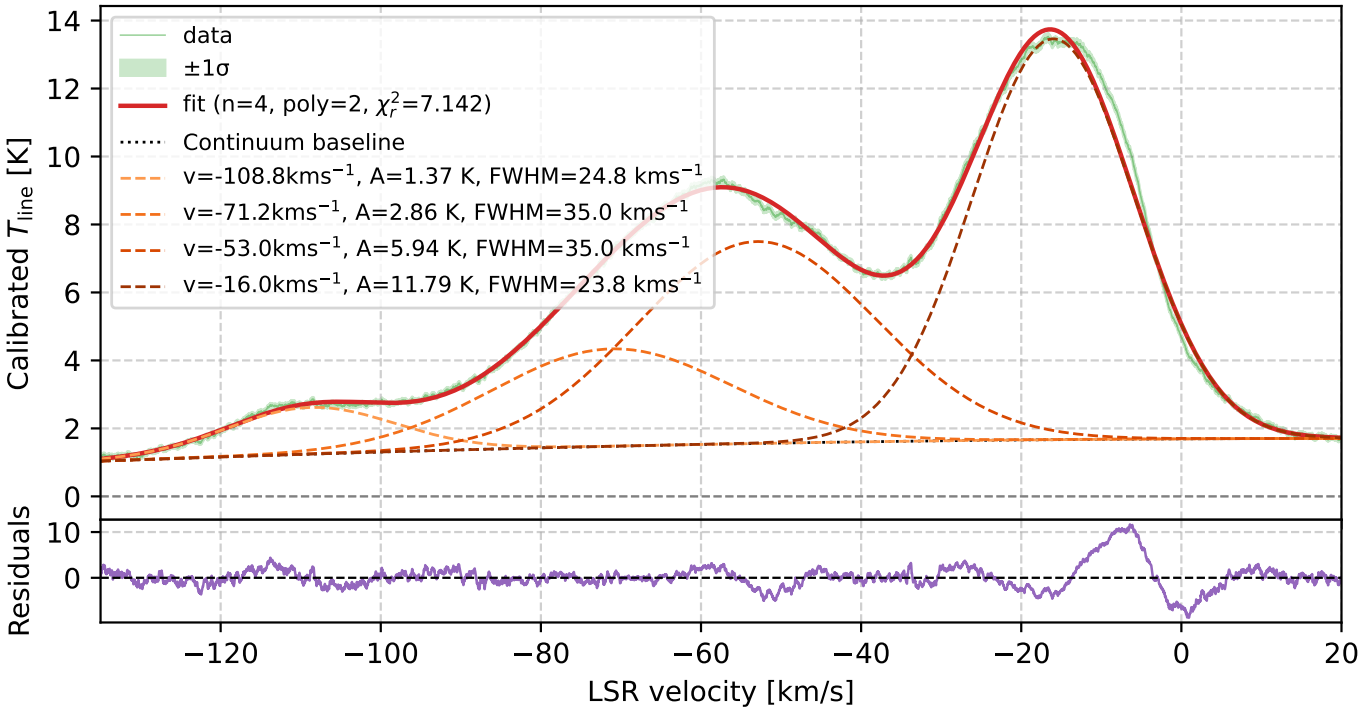

Fig. 14.— Gaussian decomposition fits for the standard field $(l, b) = (120°, 0°)$. Each panel shows the smoothed velocity profile (circles), the best-fit joint Gaussian+baseline model (solid line), individual Gaussian components (dashed), and fit residuals for the calibrated hardware-fit path. Monte Carlo perturbations of the selected fit parameters give effective profile centroid/FWHM uncertainties of about 5.6/5.2 km s^{-1} for the R branch and 0.42/0.65 km s^{-1} for $R_{\rm inv}$, indicating likely component degeneracy in the standard-field R solution.

choose the R branch as the cleaner decomposition.

7. Discussion

7.1. Receiver Noise Temperature

The measured receiver noise temperatures ($T_{\rm rx} = 79$ K at 1420 MHz and 96 K at 1421 MHz) are reasonable for a system using consumer-grade RTL-SDR components combined with low-noise amplifiers. Typical horn-antenna receiver systems span 50–300 K (Wilson et al. 2013); our values are in the lower-middle of this range.

7.2. Cable Characterization

The SDR attenuation estimate ($\alpha = 0.73$ dB/m) and power-meter estimate ($\alpha = 0.80$ dB/m) agree within 0.6σ. The standard specification for RG-58 at 1420 MHz is approximately 0.5–0.8 dB/m depending on quality, so both measurements are physically reasonable.

The unknown roof–lab cable length estimates converge on ~ 36 m from the direct methods (bench measurement and power-meter intercept), while the SDR estimate (~ 40 m) has much larger uncertainty. We adopt $L = 36$ m as our best estimate. This gives a total cable attenuation of $36 \times 0.80 \approx 29$ dB, which is compensated by the $3 \times 26 = 78$ dB of rooftop gain.

The measured $v_{\rm coax}/c = 0.54 \pm 0.10$ is about 1.2σ below the nominal RG-58 specification ($v/c = 0.66$), but remains consistent within uncertainty. This offset most likely reflects the ambiguity in identifying the periodic square-wave branch and the exact echo-edge pick.

7.3. HI Line Structure at $(l, b) = (120°, 0°)$

The standard field exhibits at least four kinematic components spanning $v_{\rm LSR} \approx -110$ to $+20$ km/s. This multi-component structure is expected for a Galactic-plane sight line in the second quadrant ($l = 120°$), which intersects multiple spiral arms:

- The Local Arm ($v_{\rm LSR} \sim -10$ to $+20$ km/s) contributes the dominant component at $v_0 \approx -16$ to $+18$ km/s depending on the extraction path.
- The Perseus Arm ($v_{\rm LSR} \sim -50$ to -80 km/s) contributes the broad components near -53 to -71 km/s with FWHM $= 35$ km/s.
- A possible outer arm feature appears near -90 to -108 km/s.

The exact velocity assignments depend on the adopted Galactic rotation model. These velocities are broadly consistent with HI4PI survey values for this sight line

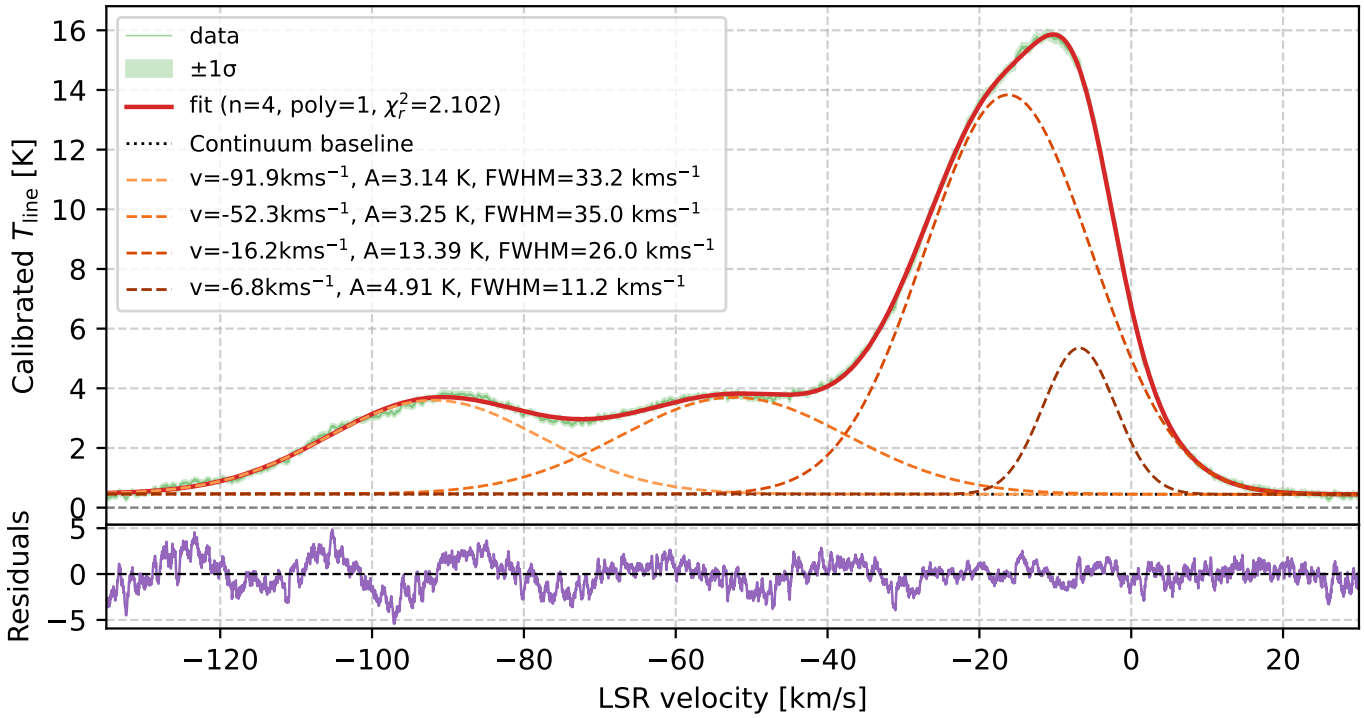

Fig. 15.— Gaussian decomposition fits for the Cygnus-X cross-validation field. Each panel shows the smoothed velocity profile (circles), the best-fit joint Gaussian+baseline model (solid line), individual Gaussian components (dashed), and fit residuals for the calibrated hardware-fit path. Monte Carlo perturbations give effective profile centroid/FWHM uncertainties of about 0.07/0.11 km s^{-1} for the selected R solution and 0.97/1.94 km s^{-1} for $R_{\rm inv}$, consistent with the visibly tighter constraint in the R branch.

(HI4PI Collaboration et al. 2016; Dickey & Lockman 1990; Kalberla & Kerp 2009). The Big Horn horn has a large field of view, and neither the effective beam size nor the beam/aperture efficiency was independently mapped in this lab. The observed components are therefore beam-averaged over a broad patch of the Galactic plane and are likely coupled to the surrounding large-scale HI structure, not just to a few discrete line-of-sight features.

7.4. Fit Quality and Model Limitations

The standard field fits show $\chi^2_{\rm red} = 6.14$–7.14 for the best 4-component models. Values $\gg 1$ indicate that the model is imperfect. This could be due insufficient number of Gaussian components where the upper bound of $N = 4$ in the grid search may not capture all kinematic sub-structure, or residual low-order baseline structure not being fully captured by the jointly fit polynomial among other likely causes. A less constrained mixture can push χ^2_ν closer to 1, but only by accepting unphysical component peak amplitudes, centroid velocities, or FWHM values that absorb baseline or calibration structure. We therefore prefer the physics-aware bounded solutions even when their reduced χ^2 remains significantly above unity.

The Cygnus-X field achieves $\chi^2_{\rm red} = 2.10$, closer to unity, suggesting that a less complex kinematic structure along that sight line is better captured by the same 4-component model.

8. Conclusion

In this lab, we performed a 3-part analysis of the Big Horn receiver.

Starting with equipment calibration, we characterized coax attenuation ($\alpha = 0.80$ dB/m, power-meter), unknown cable length ($L \approx 36$ m), coax propagation velocity ($v_{\rm coax}/c = 0.54 \pm 0.10$), SDR linearity (29 dB headroom, RMSE = 0.098 dB), and FIR+summing passband (FIR-only edge/center = 0.237, analysis band ±0.8 MHz). Next, we performed temperature calibration via Y-factor measured $T_{\rm rx} = 79 \pm 4$ K (LO 1420) and 96 ± 4 K (LO 1421), with load-temperature uncertainty dominating. System noise floors of $\delta T_{\rm chan} = 603.2$ and 697.6 mK were verified via the radiometer equation. With a more understood system, we performed HI observations at $(l, b) = (120°, 0°)$ revealed a multi-component HI spectrum spanning −110 to +20 km/s LSR, interpreted as Local Arm, Perseus Arm, and outer-arm gas. A Cygnus-X cross-validation field confirmed the ~ -16 km/s Local Arm feature. Gaussian decomposition (4 components, $\chi^2_{\rm red} = 6.14$–7.14) demonstrated the full analysis pipeline.

Table 2: Gaussian component fits for the standard field $(l, b) = (120°, 0°)$. Components with FWHM = 35 km/s hit the upper bound constraint. $\chi^2_{\rm red}$ refers to the fit over the full profile window.

Field	Comp	A (K)	v_0 (km/s)	FWHM. (km/s)
Standard	1	1.37	−108.8	24.8
R.	2	2.87	−71.2	35.0*
	3	5.94	−53.0	35.0*
	4	11.79	−16.0	23.8
$\chi^2_{\rm red} = 7.14$				
Standard	1	2.27	−90.5	28.7
$R_{\rm inv}$	2	4.67	−36.1	23.4
	3	3.04	+0.8	13.2
	4	14.16	+18.2	35.0*
$\chi^2_{\rm red} = 6.14$				
Cygnus-X	1	3.14	−91.9	33.2
R	2	3.25	−52.3	35.0*
	3	13.39	−16.2	26.0
	4	4.91	−6.8	11.2
$\chi^2_{\rm red} = 2.10$				

NOTE.—*FWHM = 35 km/s upper bound.

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Special thanks to my lab partners Nathaneil and Kayhan, and especially Mitchell and Wesley for group observations, and to Prof. Aaron Parsons and GSI Ben for instructional support. I also acknowledge the use of ChatGPT Codex and Claude Code for assistance with vetting phrasing and checking the report writeup.

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A. Reproducibility Notes

All code, notebooks, cached analysis products, and report sources used for this lab are version-controlled at `https://github.com/raw-asparagus/ay-121`.

B. Experimental Bench Configurations

This appendix provides schematic block diagrams for the distinct bench experiments in Sections 4 and 5. Node styles follow Figure 3; additional symbols (*siggen*, *powermeter*, *scope*, *termination*, *splitter*) are defined locally.

B.1. Cable Attenuation (Known-Length Cables)

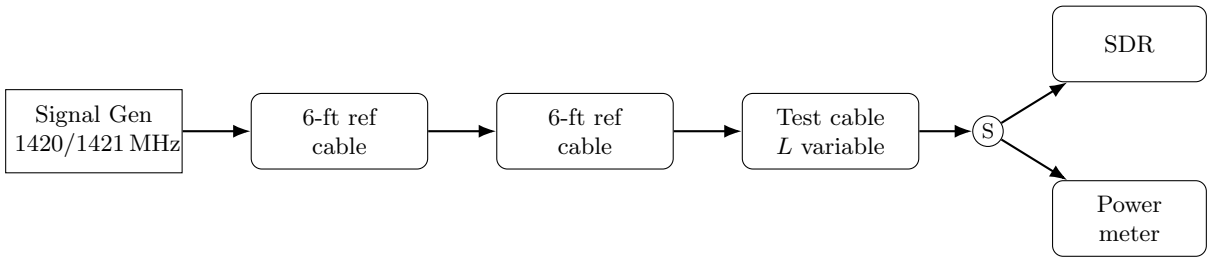

Fig. 16.— Cable attenuation bench configuration. A signal generator at 1420 or 1421 MHz feeds through two fixed 6-ft reference cables and one test cable of variable length L. A Mini-Circuits ZFSC-2-372-S+ two-way splitter at the output divides power between the RTL-SDR and a calibrated power meter, providing two independent attenuation measurements per cable run.

B.2. Unknown Roof–Lab Cable Length

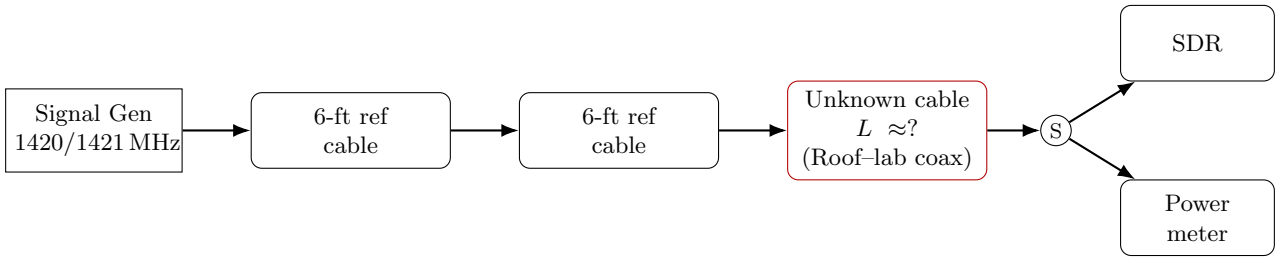

Fig. 17.— Unknown roof–lab cable length measurement. Identical to the known-cable attenuation setup (Appendix 16), including the Mini-Circuits ZFSC-2-372-S+ two-way splitter, with the test cable replaced by the unknown roof–lab coaxial cable. Three independent length estimates are derived from the SDR power reading, the power-meter reading (via intercept of the fitted attenuation model), and a bench inch-ruler (36 m).

B.3. Square-Wave Reflectometry

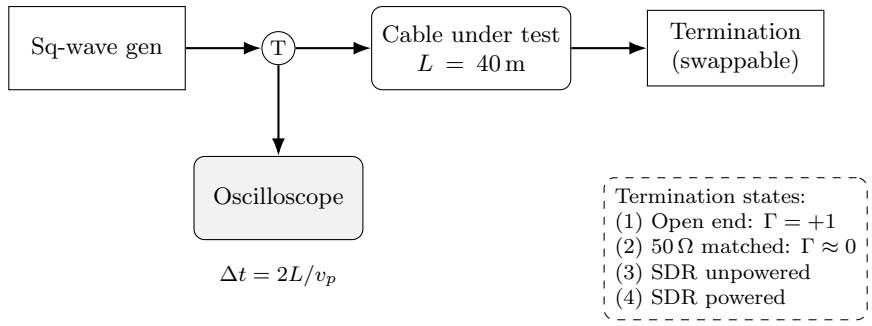

Fig. 18.— Square-wave reflectometry bench configuration. A square-wave generator is connected to a T-connector; one arm feeds the cable under test (40 m) terminated by a swappable load, and the other arm goes to an oscilloscope monitoring the round-trip echo. Four termination states were tested; the open-end state ($\Gamma = +1$) provides the clearest echo for measuring $\Delta t_{\mathrm{mod}} = 98.0\,\mathrm{ns}$; with square-wave period $T = 400.0\,\mathrm{ns}$ and branch $n = 1$, this yields $\Delta t = 498.0\,\mathrm{ns}$ and hence $v_{\mathrm{coax}}/c = 0.54 \pm 0.10$.

B.4. SDR Gain Sweep (Linearity Test)

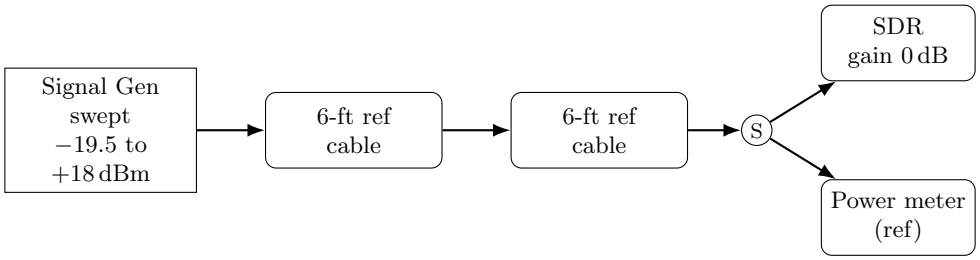

Fig. 19.— SDR gain linearity sweep. A signal generator at 1420 MHz is swept from −19.5 to +18 dBm through two 6-ft reference cables. A Mini-Circuits ZFSC-2-372-S+ two-way splitter divides power to both the RTL-SDR (fixed gain 0 dB) and a calibrated power meter providing the reference level at each setpoint. The SDR remains linear up to 13.0 dBm (RMSE = 0.098 dB), with first clipping at 15.5 dBm and 29 dB of headroom above sky levels.

B.5. Y-Factor Temperature Calibration Setup

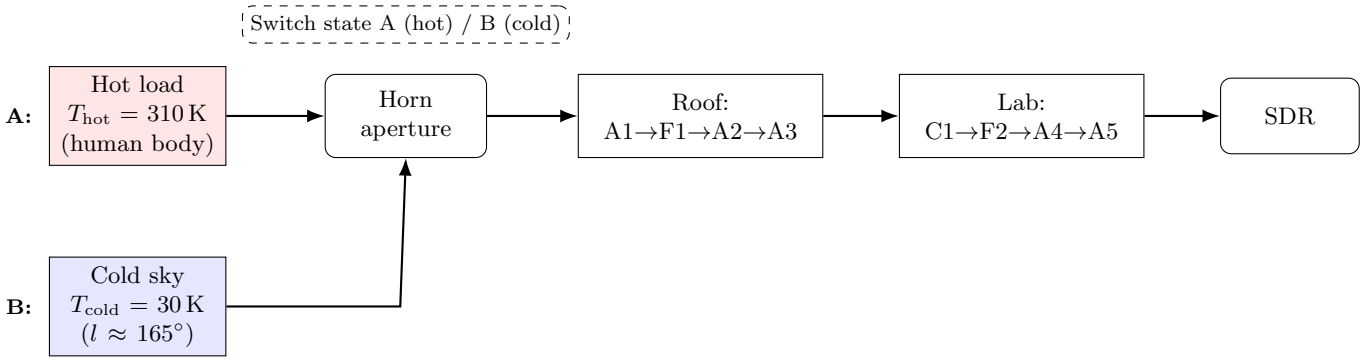

Fig. 20.— Y-factor temperature calibration setup. In state A (hot), a human body ($310 \pm 5\,\mathrm{K}$) fills the horn aperture. In state B (cold), the horn points to a low-HI column direction ($l \approx 165°$, $b \approx 36°$, $T_{\mathrm{cold}} \approx 30 \pm 2\,\mathrm{K}$). Both states use the same RF chain (roof amplifiers A1, filter F1, amplifiers A2/A3; roof–lab cable C1; lab filter F2, amplifiers A4/A5; RTL-SDR), so the Y-factor $Y = P_{\mathrm{hot}}/P_{\mathrm{cold}}$ cancels the gain and isolates T_{rx}.